RAMDEOBABA

UNIVERSITY

PROJECT TITLE

Blockchain-Secured Digital Land Registry with Cloud-Native Web Platform

- Team Members & Roll Numbers — [Enter names & roll numbers]
- Guide/Mentor Name — [Enter guide name]
- Department / School — [Enter department]
- Academic Year — 2025–26

Idea lab/Honor/ Minor/Major

COURSE CODE

[Enter course code]

PROBLEM STATEMENT

What problem are we addressing?

Land ownership still runs on fragmented paper registries and siloed digital portals. Title transfers depend on slow manual scrutiny, so records remain easy to forge and hard to verify. There is no single, tamper-proof source of truth for who owns what.

Who is affected?

Citizens and buyers risk the same plot being sold to multiple parties. Registrars drown in manual verification workloads, while banks hesitate to lend against titles that cannot be cryptographically proven.

Current challenges

Forged title deeds and duplicate sales of one property; no immutable ownership history visible to anyone; verification cycles that stretch for weeks; zero public audit trail; records scattered across departments and district offices.

Why is the problem important?

Land and property disputes drive roughly two-thirds of India civil litigation (Daksh access-to-justice study), locking property capital for decades. Secure, verifiable digital titling is a declared national priority under the DILRMP programme.

EXISTING SOLUTION / LITERATURE REVIEW

Existing methods & products

- **State land-record portals** — Bhulekh, Mee Bhoomi, Dharani digitized records, but stored in centralized, mutable databases.
- **Sub-Registrar offices** — in-person deed registration with paper annexures and wet-ink signatures.
- **DILRMP (Govt. of India)** — national programme digitizing records, maps and integration.
- **Blockchain land-title pilots** — Georgia, Sweden and Andhra Pradesh proof-of-concepts for registry records.

Limitations of existing solutions

- Centralized databases remain tamper-prone — privileged writes leave no trace.
- No end-to-end ownership provenance that a buyer can independently verify.
- Fraud surfaces only after disputes arise; nothing is screened at submission time.
- Documents verified visually — scans carry no cryptographic proof of authenticity.

RESEARCH / MARKET GAP

No existing platform unifies property registry, document integrity, transfer workflow, fraud screening and immutable audit in one always-on system. Land Ledger integrates all five layers and compares favorably on every axis — ownership truth moves on-chain, documents are SHA-256 sealed, and every action is auditable in real time.

PROPOSED SOLUTION

LAND LEDGER — a role-based web registry where every property, document and ownership transfer is cryptographically verifiable and settled on a blockchain ledger.

01

Project idea

One integrated platform for citizens and registrars: register properties, upload deed documents, initiate transfers and trace provenance — with fraud screening built into the workflow itself.

02

Proposed approach

React SPA talks to an Express REST API; MongoDB Atlas stores users, properties, documents and transfers, while a Solidity smart contract (LandRegistry.sol) holds on-chain ownership truth. Documents are SHA-256 hashed at upload; every transfer is scored by a 9-check fraud engine before approval.

03

Key innovation

The smart contract itself rejects provenance mismatches — ownership can only move along the chain of legitimate owners. Fraud is screened at submission, not after the dispute, and every API action appends to an immutable audit stream streamed live to the dashboard.

04

Expected outcome

A single trusted source of property truth, buyer-verifiable provenance, fraud blocked before settlement, and 24/7 availability from MongoDB Atlas cloud — no local database to keep running.

OBJECTIVES

1. Digitize the land registry behind secure, role-based access — separate registrar and citizen portals with JWT + bcrypt authentication.
2. Anchor every ownership change on an immutable EVM ledger through the LandRegistry.sol smart contract with on-chain provenance enforcement.
3. Guarantee document integrity using SHA-256 hashing at upload with independent re-verification on demand.
4. Automate transfer risk screening with a deterministic 9-check fraud engine (PASS / WARN / FAIL) before registrar approval.
5. Ship a production-grade platform with CI/CD pipelines, Prometheus + Grafana observability and 24/7 MongoDB Atlas cloud storage.

100%	0	1	29/29	0
transfers settled on-chain	unverified documents accepted	audit event per API action	API · 20/20 contract tests	local DB dependency (Atlas M0)

SYSTEM ARCHITECTURE / METHODOLOGY

CLIENTS

- **Citizen Portal** — React 18 · Vite · Tailwind CSS (responsive 360–1280 px)
- **Registrar Console** — React 18 · Vite · Tailwind CSS (responsive 360–1280 px)

API LAYER

- **Express REST API** — Helmet · JWT + bcrypt auth · RBAC · rate limiting · /health · /metrics

CORE SERVICES

- **Fraud engine** — 9 deterministic checks
- **Document service** — SHA-256 + Multer
- **Transfer orchestrator**
- **Audit service**

DATA & LEDGER

- **MongoDB Atlas** — 8 collections · 24/7 cloud
- **LandRegistry.sol** — local EVM via Hardhat · ethers v6
- **Uploads volume** — original PDFs

END-TO-END TRANSFER WORKFLOW

01 Register property → 02 Upload & hash documents → 03 Initiate transfer → 04 Fraud screening → 05 Registrar approval → 06 On-chain settlement → 07 Audit event logged

IMPLEMENTATION

Frontend

React 18 · Vite · Tailwind CSS · light/dark themes · responsive to 360 px

Backend

Node.js · Express · Helmet · JWT + bcrypt · Multer uploads

Database

MongoDB Atlas M0 · Mongoose 8 · atomic counters · 8 collections

Blockchain

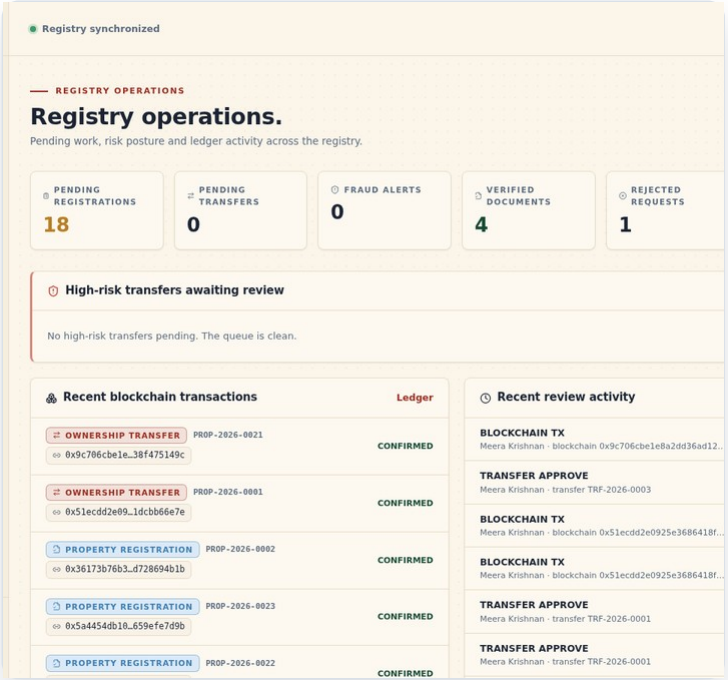
Solidity 0.8.24 · LandRegistry.sol · Hardhat · ethers v6

DevOps

Docker · docker-compose · Jenkins (10-stage) · Kubernetes · Prometheus + Grafana

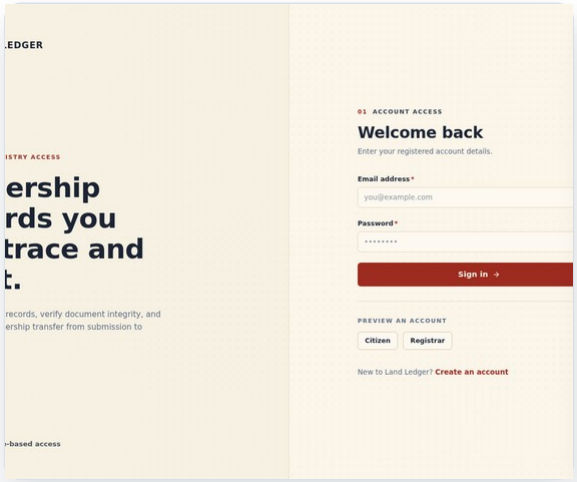
Testing

supertest (29 API tests) · Hardhat toolbox (20 contract tests) · browser E2E sweep

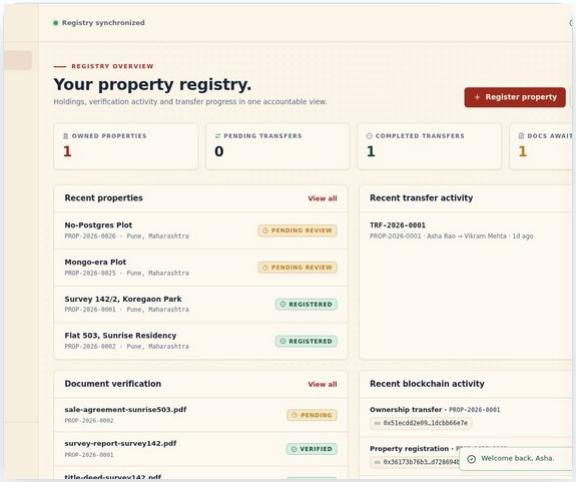


Live registrar console — real ownership-transfer settlements on the local chain (block 15, tx 0x9c706cbe...)

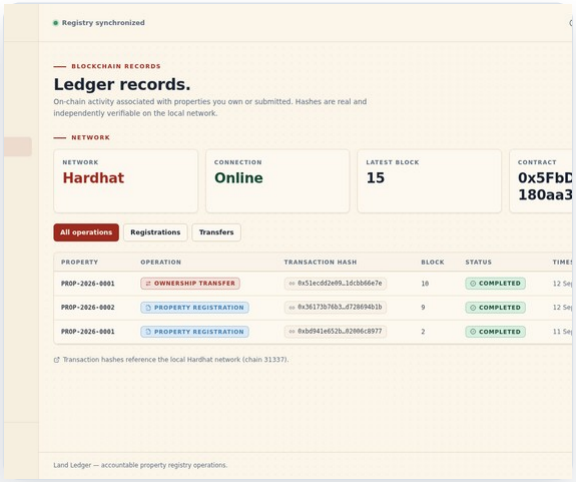
RESULTS / PROTOTYPE



Secure sign-in — role-based access



Citizen dashboard — owned properties & transfers



Public ledger — every on-chain transaction

✓ VERIFIED RESULTS

29/29
backend API tests passing

20/20
smart-contract tests passing

29/29
end-to-end demo checks

Real tx settled
block 15, 0x9c706cbe…

0
overflow 360–1280 px viewports

Health green
on Atlas alone — PostgreSQL fully removed

INNOVATION, IMPACT, DELIVERABLES & FUTURE SCOPE

Innovation

- On-chain provenance enforcement — contract rejects ownership-chain mismatches
- Fraud screened at submission time via 9 deterministic checks
- SHA-256 tamper evidence for every deed document
- Live immutable audit stream from Atlas on the registrar dashboard

Impact

- Buyers independently verify ownership before paying
- Registrars decide faster with risk scores pre-computed
- Courts receive immutable, timestamped evidence
- Registry reachable 24/7 from anywhere — zero local infrastructure

Deliverables

- Full-stack web platform — 45+ frontend files, 10 API modules
- Audited LandRegistry.sol contract with 20-test suite
- REST API verified by 29 automated tests + E2E browser sweep
- Docker / Kubernetes / Jenkins CI-CD + Grafana observability

Future scope

- Hosted testnet deployment (Polygon Amoy) for public access
- Aadhaar-based e-KYC for identity assurance
- GIS map layers and OCR auto-extraction of deed data
- SMS / WhatsApp alerts and AI-assisted valuation hints

CONCLUSION & REFERENCES

CONCLUSION

Land Ledger proves that trustworthy digital land administration can be built end-to-end with open technology.

- ✓ Every record hashed, every transfer fraud-screened before approval
- ✓ Ownership settled on an immutable ledger with full provenance
- ✓ Every action audited — available 24/7 from the cloud with nothing more than a browser and an internet connection.

References

1. Department of Land Resources, Govt. of India — Digital India Land Records Modernization Programme (DILRMP).
2. Daksh — Access to Justice surveys on land/property share of civil litigation in India.
3. Ethereum Foundation — Solidity 0.8.x language documentation.
4. Nomic Foundation — Hardhat and ethers.js v6 developer documentation.
5. MongoDB Inc. — MongoDB Atlas M0 and Mongoose ODM v8 documentation.
6. Meta Open Source — React 18, Vite and Tailwind CSS documentation.
7. V. Lemieux — Evaluating the Use of Blockchain in Land Registries, 2019.